

Question	Answer	Marks
3(a)	work done per unit time	B1
3(b)(i)	$P = F \times v$ $= 1750 \times 35$	C1
	$= 6.1 \times 10^4 \text{ W}$	A1
3(b)(ii)	$W = Fs$ $= 1750 \times 17\,000$	C1
	$= 3.0 \times 10^7 \text{ J}$	A1
	or $W = Pt$ $= 6.1 \times 10^4 \times (17\,000 / 35)$	(C1)
	$= 3.0 \times 10^7 \text{ J}$	(A1)

Question	Answer	Marks
3(b)(iii)	$P = V \times I$ Power in = $600 \times I$	C1
	Efficiency = $\frac{\text{useful power output}}{\text{(total) power input}}$	C1
	$0.85 = 6.1 \times 10^4 / (600 \times I)$ $I = 120 \text{ A}$	A1
3(c)(i)	Air resistance is the same, as the speed is the same	B1
3(c)(ii)	The motor is producing less power (because of gravitational force / conversion of gravitational potential energy to kinetic energy) so the current will be smaller.	B1

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